ACT SAT GO
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SAT MATH

Formulas the SAT won't give you

The reference sheet skips every one of these.
9 formulas inside.

$$(x-h)^2+(y-k)^2=r^2$$

$$x=\frac{-b}{2a}$$

$$b^2-4ac$$

save this. seriously.

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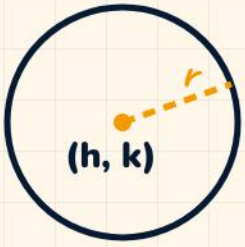
01 FORMULAS THE SAT WON'T GIVE YOU

Equation of a circle

$$(x-h)^2+(y-k)^2=r^2$$

WHAT IT TELLS YOU

Center is **(h, k)**. The number on the right is **r²**, so the radius is its square root.



Given $x^2+y^2-6x+4y=12$? Complete the square: $(x-3)^2+(y+2)^2=25$, so center (3, −2), radius 5.

Watch out: (x + 3)² means h = −3. The sign flips!

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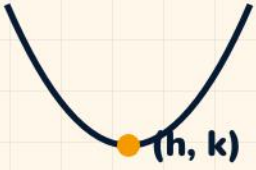
02 FORMULAS THE SAT WON'T GIVE YOU

Vertex form of a parabola

$$y = a(x - h)^2 + k$$

WHAT IT TELLS YOU

The vertex is **(h, k)**. If **a > 0** it opens up and k is the minimum; if **a < 0** it opens down and k is the maximum.



Example: $y = 2(x - 3)^2 + 1$ has vertex (3, 1) and a minimum value of 1.

Watch out: y = (x + 5)² − 2 has its vertex at (−5, −2), not (5, −2).

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03 FORMULAS THE SAT WON'T GIVE YOU

Find the vertex in 5 seconds

$$x = \frac{-b}{2a}$$

WHAT IT TELLS YOU

For $y = ax^2 + bx + c$, this gives the x-value of the vertex. Plug it back in to get the max or min.

Example: $y = x^2 - 8x + 3 \rightarrow$ x = 4, and y = 16 − 32 + 3 = −13. Vertex (4, −13).

Watch out: asked for the maximum VALUE? That's the y, not the x.

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04 FORMULAS THE SAT WON'T GIVE YOU

Sum and product of the solutions

$$\text{sum} = \frac{-b}{a} \qquad \text{product} = \frac{c}{a}$$

WHAT IT TELLS YOU

For **$ax^2 + bx + c = 0$** you get the sum and product of the solutions without solving.

Example: $2x^2 - 8x + 3 = 0 \rightarrow$ sum = 4, product = 3/2.

Watch out: move everything to one side and set it equal to 0 first.

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05 FORMULAS THE SAT WON'T GIVE YOU

The discriminant

$$b^2 - 4ac$$

WHAT IT TELLS YOU

Tells you how many real solutions: **positive** = 2, **zero** = 1, **negative** = 0.

Example: $x^2 + 4x + 4 \rightarrow$ 16 – 16 = 0, so exactly one solution.

Use it when they ask how many times a line hits a parabola: set equal, move to one side, check $b^2 - 4ac$.

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06 FORMULAS THE SAT WON'T GIVE YOU

Percent change

$$\frac{\text{new} - \text{old}}{\text{old}} \times 100$$

WHAT IT TELLS YOU

Always divide by the **original** amount, not the new one.

$40 → $50 is a **25% increase**. $50 → $40 is only a **20% decrease**.

Watch out: up 20% then down 20% does NOT get you back. 100 → 120 → 96.

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07 FORMULAS THE SAT WON'T GIVE YOU

Exponential growth and decay

$$y = a(1 \pm r)^t$$

WHAT IT TELLS YOU

a = starting amount, **r** = rate as a decimal, **t** = time. Growing? Use 1 + r. Shrinking? Use 1 − r.



Losing 15% a year means multiplying by **0.85** every year.

Watch out: "doubles every 3 years" is a·2^(t/3), not a·2^t.

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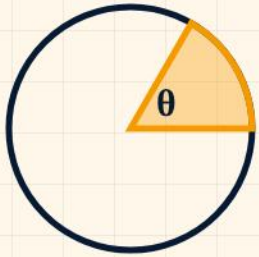
08 FORMULAS THE SAT WON'T GIVE YOU

Arc length and sector area

$$\text{arc} = \frac{\theta}{360} \cdot 2\pi r \qquad \text{sector} = \frac{\theta}{360} \cdot \pi r^2$$

WHAT IT TELLS YOU

θ/360 is the slice of the circle you have. Take that slice of the circumference or the area.



Example: r = 6, θ = 60° → arc = 2π, sector = 6π.

Watch out: angle in radians? Then arc = rθ.

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09 FORMULAS THE SAT WON'T GIVE YOU

Interior angles of a polygon

$$\text{sum} = (n - 2) \cdot 180°$$

WHAT IT TELLS YOU

n is the number of sides. Divide by n to get each angle of a regular polygon.



Hexagon: (6 – 2) · 180 = 720° total, so each angle is 120°.

Bonus: the exterior angles of ANY polygon always add up to 360°.

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SAT MATH

Save this for test day.

Which one did you NOT know? Tell us in the comments.

- Circle
- Vertex form
- Vertex shortcut
- Sum & product
- Discriminant
- Percent change
- Growth & decay
- Arc & sector
- Polygon angles

Want these explained 1-on-1? DM us FORMULAS

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